

Joint tests and interpretation

Slide route: Lecture6v2.tex: single coefficients, joint hypotheses, F tests, linear restrictions and confidence sets. Chapter 7.

Session 8 | A claim about several coefficients

Outcomes: Distinguish separate from joint tests, calculate a classical F statistic, test a linear combination, and interpret restrictions in context.

100-minute route

0–10	Partial-coefficient retrieval.
10–25	Individual versus joint hypotheses.
25–45	F-statistic and restricted/unrestricted example.
45–55	Robust versus classical tests and pause.
55–75	Single restrictions on multiple coefficients.
75–90	Confidence sets and model specification.
90–100	Exit ticket and functional-form preview.

Worked example | Two exclusions

Use with slides Lecture6v2 slides 7–16: Tests of Joint Hypotheses through the example on *strs* and *frpmfracs*. Use the SSR calculation when slide 11 introduces the F-statistic, then compare it with the deck’s example on slide 16.

Test $H_0 : \beta_2 = \beta_3 = 0$. A restricted regression has residual sum of squares 120; the unrestricted regression has 100. Let $n = 100$, unrestricted slope count $k = 3$, and restriction count $q = 2$.

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n - k - 1)} = \frac{(120 - 100)/2}{100/96} = 9.6.$$

Under homoskedastic normal errors, compare with $F_{2,96}$; its 5% critical value is about 3.09, so reject.

What to say The joint claim that both coefficients are zero is inconsistent with the data under these assumptions. This does not establish that each individual coefficient is nonzero.

Robust caveat The residual-sum-of-squares formula is the homoskedastic-only test. Under heteroskedasticity, use the estimated robust covariance matrix in a Wald test; do not relabel this formula robust.

Worked example | Equality is one restriction

Use with slides Lecture6v2 slides 22–24: Testing Single Restrictions, Rearranging, and Direct Testing. Use this equality example before the Stata command so students see why the covariance term matters.

Suppose $\hat{\beta}_1 = .5$, $\hat{\beta}_2 = .2$, their estimated variances are .04 and .01, and covariance is .01. To test $H_0 : \beta_1 = \beta_2$:

$$\widehat{\beta_1 - \beta_2} = .3, \quad SE(\hat{\beta}_1 - \hat{\beta}_2) = \sqrt{.04 + .01 - 2(.01)} = \sqrt{.03}.$$

$$t = .3/\sqrt{.03} \simeq 1.732, \quad 95\% \text{ large-sample CI: } .3 \pm 1.96\sqrt{.03} = [-.0395, .6395].$$

Fail to reject equality at 5%. Each coefficient's separate significance relative to zero cannot answer whether the two coefficients differ.

Pair prompt Why can we not subtract the two standard errors? Because the variance of a difference includes both variances and their covariance.

Multiple-testing side example If two tests were independent and each used 5%, the probability of at least one false rejection under the joint null would be $1 - .95^2 = .0975$. Regression coefficient tests need not be independent, but this shows why separate tests do not implement a joint 5% test.

Software illustration With an appropriately fitted Stata model:

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test x2 x3
test x1 = x2
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Use the model's robust covariance estimate when robust inference is required. Confirm variable names before demonstrating.

Confidence sets A joint confidence ellipse incorporates coefficient covariance. It is not obtained by declaring the Cartesian product of individual 95% intervals a joint 95% region.

Exit answers For one restriction, a consistently constructed Wald F equals the corresponding t^2 . Equality of two coefficients is one restriction. An insignificant individual test does not preclude joint significance.

If short / preparation Leave repeated output screenshots for reading. Preserve both worked examples. Read Chapter 8 and ask students to sketch a relationship whose slope changes with X .

Source: Lecture6v2.tex; synthetic numerical examples.